

Maximising AI for driver attention and vehicle safety

Artificial Intelligence (AI) is revolutionising vehicle safety by improving driver attention and reaction to road conditions. It offers nuanced insights into driver behaviour, enhancing overall road safety.

Understanding driver reactions: AI decodes drivers' responses to various road situations, including emotional states and adherence to traffic rules.

Addressing unexpected obstacles: AI distinguishes between momentary distractions and emergencies, refining its approach to driver attention.

Real-time collision detection: AI predicts potential collisions by analysing data continuously, enhancing proactive safety measures.

Key factors in risk assessment: AI assesses variables like vehicle speed, obstacle proximity, and driver alertness for accurate risk evaluation.

Versatile adaptation: AI adapts across scenarios, from lane-keeping assistance to using drones for traffic condition monitoring.

The implementation of AI in vehicular systems marks a significant stride in automotive technology, drastically improving safety and driver awareness. Through real-time analysis and adaptive responses, AI is setting new benchmarks in vehicle safety, making driving a more secure and responsive experience.

Data storage and integration:

Using technologies like Kafka, the project can store large amounts of data for extended periods, integrating with multiple technologies for comprehensive data management.

Camel case integration layer: This layer plays a pivotal role in integrating messaging protocols with the software, ensuring seamless communication and data flow.

The Bobby Car project, with its innovative use of technology and software, is a beacon in the automotive industry, demonstrating how connected and software-defined vehicles can significantly enhance operational efficiency and user experience in smart cities, smart factories, and beyond.

Efficient real-time data processing and development framework

An essential aspect of Bobby Car's approach is the real-time management and processing of data. The system operates within the OpenShift platform, where events are stored in memory for quick access and faster processing. This setup allows for efficient data handling, essential for applications like connected cars, where GPS coordinates, metrics, and zone changes are continuously sent to the cloud. The cloud processes this data, and

the results are displayed on a dashboard, providing real-time insights necessary for decision-making or AI integration.

Bobby Car has implemented a unique software development framework called 'Tablespaces.' This approach allows developers to work without installing dependencies on their devices. They use a browser-based integrated development environment (IDE), facilitating seamless development and centralised, secure storage of code. This method streamlines the development process, allowing for efficient compilation and testing directly within the browser.

Bobby Car's development model is twofold. The first part focuses on AI and ML, crucial for leveraging the vast data from IoT devices. This pipeline involves preparing the data, developing models on top of it, and deploying these models on the unified OpenShift platform. The second part concerns application development, where developers create or modify code, pass it through a CI/CD pipeline for validation, and then deploy it on a platform compatible with various cloud environments.

The system's UIs and dashboards enable comprehensive monitoring and management of data and applications. The real-time dashboards are crucial

for observing model outputs and their applications. Furthermore, the data is stored in a time-series database, making it easier to retrieve specific data points for automated actions using AI and ML.

Bobby Car's pipeline undergoes multiple unit and quality checks, ensuring that the final software is robust and reliable. The pipeline's efficiency is evident in its operational state, allowing for a seamless transition from development to deployment.

The evolution and integration of software technologies across various industries, as exemplified by the Bobby Car project, mark a significant shift in the business landscape. The transformation driven by digital interfacing and the incorporation of cutting-edge technologies like 5G, AI, and IoT is not merely a trend but a fundamental change in how businesses operate and interact with their customers.

The Bobby Car project, in particular, stands as a testament to the potential of software-defined innovation in bridging the gap between digital and mechanical engineering. It showcases the importance of agility, adaptability, and continuous learning in developing applications that meet modern safety standards, efficiency needs, and customer expectations.

As industries move forward in this software-centric era, the lessons learned from such innovative projects will be crucial in shaping a future where technology enhances every aspect of business operation, from customer interaction to product development and service delivery. The journey ahead, filled with challenges and opportunities, is poised to redefine industry norms and set new standards for excellence in a digitally interconnected world. **END** 

This article is based on a tech talk given by Vinay Rajagopal, ISV Technology Lead, RedHat, at EFY Expo 2023 in Pune. It has been transcribed and curated by Nidhi Agarwal, Technology Journalist at EFY

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